LONDON METROPOLITAN UNIVERSITY

islington college
(इस्लिङ्टन कलेज)

Module Code & Module Title

CS6P05NI Project

Proposal

5% Individual Coursework

Submission: Final Submission

AY 2025 2026

Academic Semester: Autumn Semester 2025

Credit: 30 Credit Year Long Module

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Assignment Due Date: Wednesday, December 3, 2025

Assignment Submission Date: Wednesday, December 3, 2025

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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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1 . Introduction

Philanthropic activities are increasingly gaining prominence in Nepal, where philanthropy is already a well-established culture due to its strong Sewa, or tradition of charity, and community support in the country. Due to the recent natural disasters, such as the 2015 earthquake and the covid-19 pandemic, local and diaspora philanthropy has grown lately. How ever, the system of delivering donations in Nepal remains very fragmented and embedded with distrust. The problems of the absence of transparency, the challenge of ensuring the legality of various local NGOs and local schools, and the absence of a reliable system to oversee the donation specific to the well-intentioned ones (Bhandari, 2022).

Donors often struggle to find the reputable organizations, educational establishments or neighbourhood projects which require support or help. Owing to weak or absent updates on progress or use of funds on a project, several benevolent contributors are left wondering what happens to their contribution. This incompliance prevents further involvement because it creates an atmosphere of doubt and suspicion (Prakash, 30 March 2010).

The problem that will be addressed in this project will include the development of a single transparent donation management system. The technology will bring together legitimate NGOs and access far much information about their needs and grant funders real time tracking of their funds. . The answer that the platform offers in a well formatted technologies platform, will have an effect of increasing the credibility, performances, and accessibility of the philanthropic giving to individuals and organizations who desire to make noteworthy contributions (Charities Aid Foundation, 2023).

The project aims at reviving the element of confidence, transparency and bringing about a more stable line of philanthropic donations in the country by addressing the week areas that make the donation ecosystem in Nepal quite fragmented.

1.2 Problem scenario

It is difficult to make sure contributions as donors do not find the charitable giving in Nepal very dispersed, unorganized, and opaque. Although a significant number of citizens are eager to give to local charities, schools, and non-governmental organizations, they are often unable to locate credible organizations, verify the use of their donations, or learn about the outcomes of their donations. This information void is a significant obstacle to trust as donors are reluctant to make long-tern or recurrent donations dues to the lack of this information (Shrestha, 2019).

Most of the NGOs in Nepal still use word-of-mouth, social media updates and manual reporting, which do not provide a cohesive, verifiable, or consistent account of donor activities. This results in a reduced participation in sustainable donating (Bhandari, 2022).

Principal Obstacles

- Donors have a hard time finding credible and well-researched NGOs or educational institutions.
- There is not a lot of success in terms of making the information about how the donations are spent transparent.
- Donors are hardly sent post-donation feedback or progress reports, or updates of what has been accomplished through their donation.
- Inefficient operation and misleading information is bound to occur due to the absence of one open ecosystem that links donors, NGOs, and community activities.

Supporting Data

- Openness promotes confidence of donors

The international studies have demonstrated that there is a much better possibility of a donor to give the donation when the charities present validated impact reports and financial transparency. The long-term giving and the assurance of the donors are directly related to transparent reporting and transparency (Charities Aid Foundation, 2023).

- The trustworthiness to the donor is reduced due to the misrepresentation of information

Many members of the population are not willing to donates as they are ignorant of how their funds are being misused by the NGOs (governance studies conducted in Nepal).This is a deficiency in clarity that deters engagement and puts obstacles on philanthropic contribution (NEPAL ADMINISTRATIVE STAFFCOLLEGE, 2018).

- Insufficiency of appropriate communication separates the inputs and the outputs.

Development organisations reports and in the context of reports, there are indications that ineffective reporting systems and intermittent nature of news that NGOs feed on leave donors in a sense of disconnection with the real effects. (Bhandari, 2022).

- The Nepal NGO sector is opaque irrespective of its size.

Nepal has tens of thousands of registered Non-Governmental Organizations which have little to no transparency, accountability or disclosure. Donors have a challenge in identifying credible organizations (Social Welfare Council Annual Report, 2022).

1.3 Project as a solution

This website brings donation into one convenient platform by offering:

- The unified platform of the system also provides convenient access to trustworthy and accredited non-governmental organisations, and community organizations by the donors.

- The software is also completely transparent regarding the usage of the donations, as it regulates the whole donation cycle through the addition of the receipts, evidence, time and place data.

- The donors will never be in the dark after contributing because they receive the logs of the activities, visuals and real time updates of the progress.

- It is with well-defined requirements and up to date programs that donors can be able to access dependable, systematic and valid information at the NGOs prior to donating.

- The reason behind it is that a tough vetting process will be conducted to make sure that only real and credible NGOs and institutions will be allowed appearing on the platform.

- The transparent reports of use such as bills, before and after photos and indicators of completion help easily check financial and operational activities.

- This increases trust, concept of access and transparency to the entire system, charities giving is more dependable, substantial and responsible.

2 Project Aim and objectives

Aim: To develop a web and mobile platform that will lead to centralisation and enhance the donor's involvement, efficiency, and transparency in the process of donating to charities in Nepal.

Objectives

Specific: To develop and deploy a secure donor portal with options od a one time and recurring Donation, registering as a user, and maintaining a profile.

Achievable: To create a special platform of NGOs, which will help them to enlist their needs, build verified profiles, and will give at least two updates on the progress of each funded project.

Realistic: Add an interactive map that will enable contributors to search and filter at least thirty certified groups by their location and cause using the Google Maps API.

Time-bound: By the conclusion of the development process, the prototype should incorporate a gamification system (badges, points, and a scoreboard) and be tested in terms of user engagement.

3. Expected Outcomes & Deliverables

- **Deliverables:**
  - A fully functional and deployed web application.
  - A responsive donor portal with a personalized dashboard.
  - A comprehensive admin control panel.
  - A dedicated NGO portal.
  - An interactive map interface.
  - A secure payment integration (e.g., Khalti).
  - Project documentation (Source code, User manuals, Final Report).
- **Features:**
  - Donor profiles, donation history, and impact summaries.
  - Real-time donation tracking and progress bars for projects.
  - Gamification elements (Badges, Points, Leaderboard).
  - NGO verification and project update system.
  - Secure payment processing.

4 . Project Risks, Threats & Contingency Plans

- **Risk: Difficulty of the payment gateway integration**

  Threat: The donations process could be disrupted by the malfunctions of payments, sluggish transactions of the mistakes in the API.

  Plan of Action: By using high quality documented APIs and initiate integration of gateways at the latest.

- **Risk: Difficulties with the live server deployment.**

  Threat: Server breakdowns or Unsuccessful deployments

  Contingency Plan: Test the deployment process in a staging environment before going live. Keep backup configurations and versions of the project files.

- **Risk: API or integration issues that were caused by delays due to troubleshooting.**

  Threat: Unexpected file uploading problems, payment API, or authentication may slow down the development process and affect the development timeline.

  Contingency Plan: Allow time on testing and troubleshooting on a per-sprint basis.

5 Methodology

The Donation Management System (DMS) was planned, designed, created and tested through an organised process of Software Development Life Cycle. SDLC gives a strict framework which facilitates each stage of the project to be done in a systematic way to assist both requirement analysis and iterative development (Sommerville, 2016).

5.1 Considered methodology

a. **Prototype Methodology**

The prototype approach was firstly regarded due to the opportunity to develop the first drafts of the system elements and receive the feedback of the users prior to the complete development (Sommerville, 2016). Highlight that prototyping is essential in getting to know the expectations of the users. In the case of the DMS, prototypes might be useful in visualising dashboards, listing, and verification workflows by the NGOs and donors. This method was however not chosen because it was time consuming. Creating and optimizing various prototypes, e.g. payment flows, tracking systems, or NGO verification would eat a large portion of the already constrained project timeline.

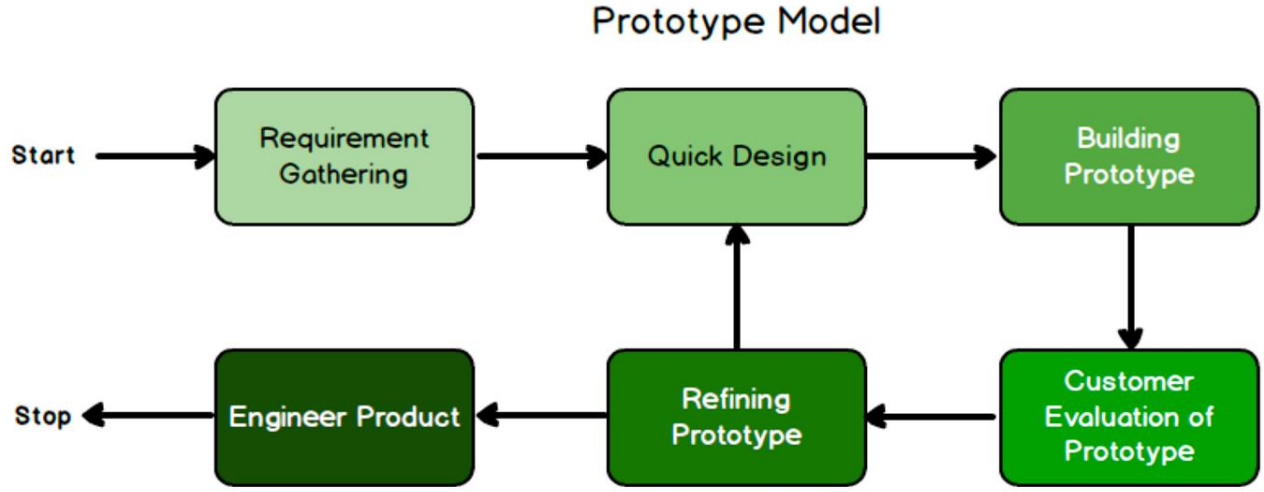

Figure 1 methodology

b. **Spiral methodology**

Spiral approach was considered because it focuses on the analysis of risks, development in a loop, and continuous improvement. Its systematic cycles of planning, prototyping and

assessment can be used in complex features such as secure payment integration and gift monitoring.

But to a single developer with minimal time on hand, the high number of review processes and the extensive documentation required in the Spiral model can be time consuming. Agile Scrum is a more flexible and efficient approach to rapid development, although dealing with constant risk evaluation and planning would hamper the process (Boehm, 1988).

5.2 Used Methodology

Agile Methodology (Scrum Framework)

Agile Scrum method was chosen since it advocates flexibility, constant user feedback and incremental development that is most suitable in a highly dynamic system such as a DMS. Agile opens the way to flexible planning, constant improvement, and, therefore, a single developer will be effective in adapting to project changes (Dingsøyr, 2012).

Scrum breaks down the work to a series of short sprints, the outcomes of which are operating elements, such as payment integration, donor verification, NGO onboarding, or real-time updates. This is a systematic cycle in which evaluation and improvement can be undertaken periodically based on the requirement of its users.

Benefits of Scrum

- Incremental - Scrum assists in delivery of working modules after each sprint, which improves final outcomes in terms of quality through refinement.
- High Flexibility – Agile allows introducing new requirements into the flow of development and does not disrupt the process of development.
- User Centred Approach - A review is conducted regularly to ensure that the donor, NGO, and supervisor feedback is present in every version.
- Improved Productivity: The workload pressure is reduced as the tasks are divided into manageable parts, and the steady progress is ensured.
- Reduced Risks: The high frequency of testing in every sprint is used to detect the issues early and prevent major failures in the making of the development process.

6. Resource requirement.

6.1 Hardware requirements

Computer:

A personal laptop is required to develop and test the Donation management System. It should have enough processing, power, memory, storage to run and deploy the system. and it should be able to handle multiple services smoothly.

Internet Connection:

A stable high-speed internet connection is necessary for accessing online libraries, APIs, documentation and version control platforms.

6.2 Software Requirements

- **IDE:** Visual Studio Code
- **Frontend Framework:** React.js
- **Backend Runtime Environment:** Node.js
- **Backend Framework:** Express.js
- **Database:** MongoDB
- **Version Control:** GitHub
- **API Testing Tool:** Postman

7.Work Breakdown Structure

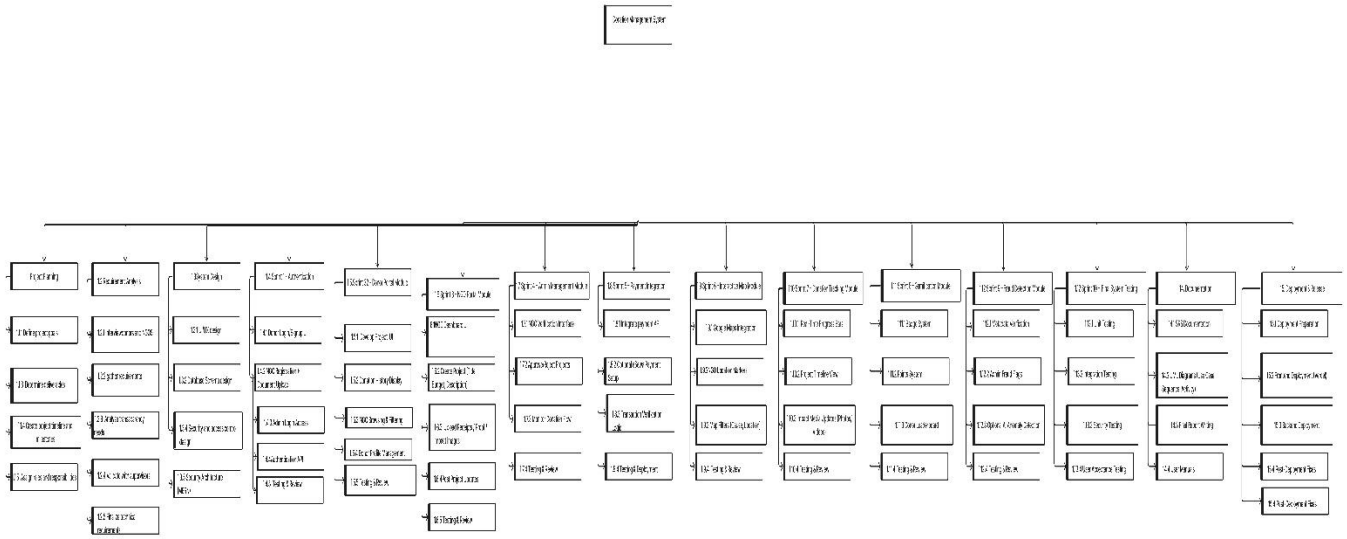

Figure 2 Work Breakdown Structure

8 Milestone

Milestones are a significant element during the project since they serve as a check point and they could be applied to analyse the progress of the development, pinpoint delays, and define the general state of the development process.

Project Initiation:

Create project scope, critical user roles, develop initial UI/UX wireframes and develop the database schema.

Sprint 1: Sprint will install and test donor, NGO and admin authentication modules.

Sprint 2: Creation of donor dashboard with NGO browsing and donation history.

Sprint 3: Prepare NGO portal on project creation and project verification uploading.

Sprint 4: The administrator panel that will verify the NGOs and check the project submissions.

Sprint 5: Introduce payment gate way

Sprint 6: Implement interactive maps as the feature

Sprint 7: monitoring bars of donations and impact updates

Sprint 8: Gamification (make badges, points)

Sprint 9: Detection instruments Author fraud detection, e.g. metadata checks and admin flags.

Sprint 10: Complete system testing security and user acceptance testing.

Documentation: Write up all project documentation, e.g. the SRS, design diagrams, final report, testing logs and user manuals.

Deployment: Install the hosting environment, undergo deployment tests and install the final version of the platform that is production ready.

9. Gantt Chart

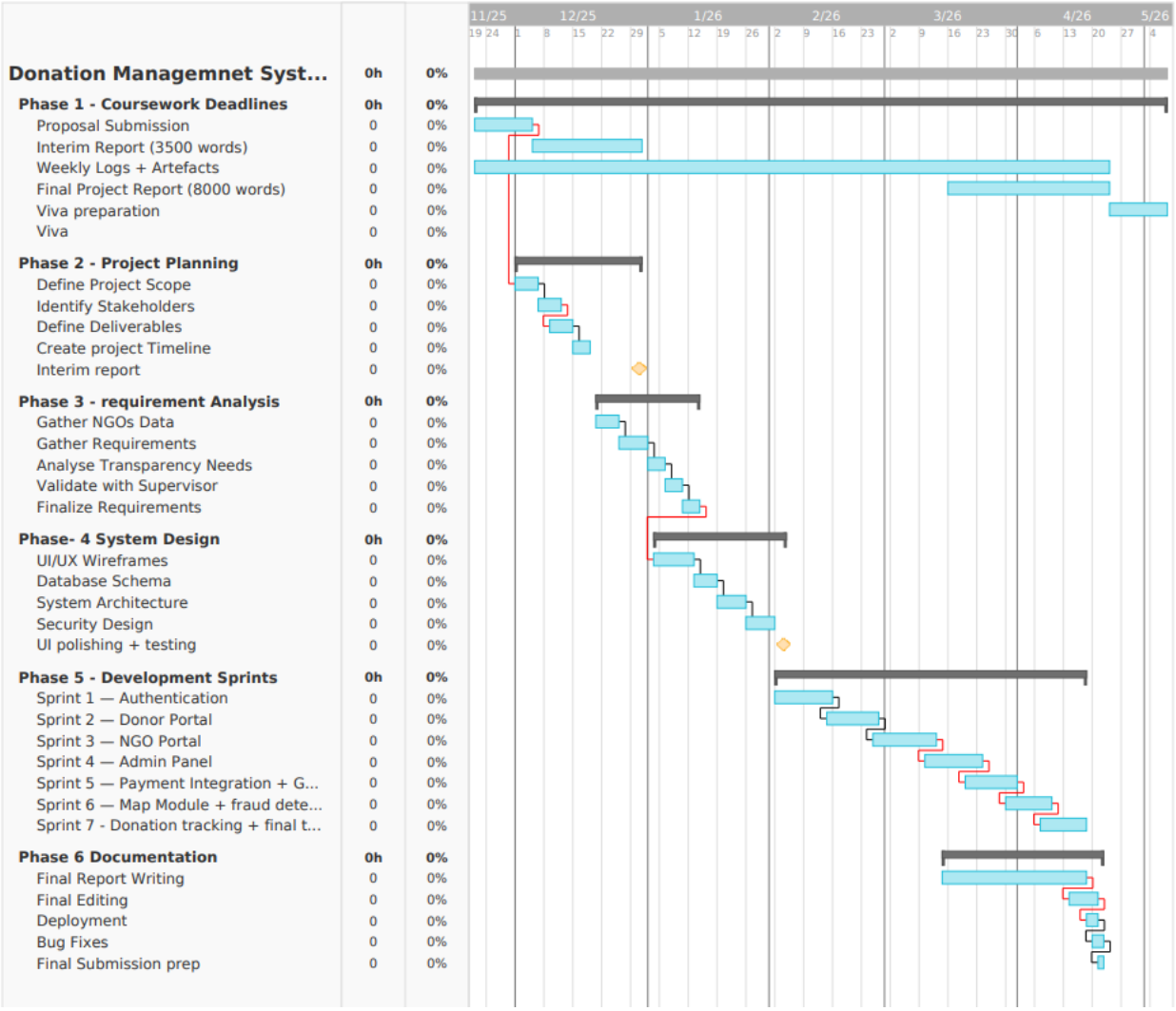

Figure 3 Gantt chart

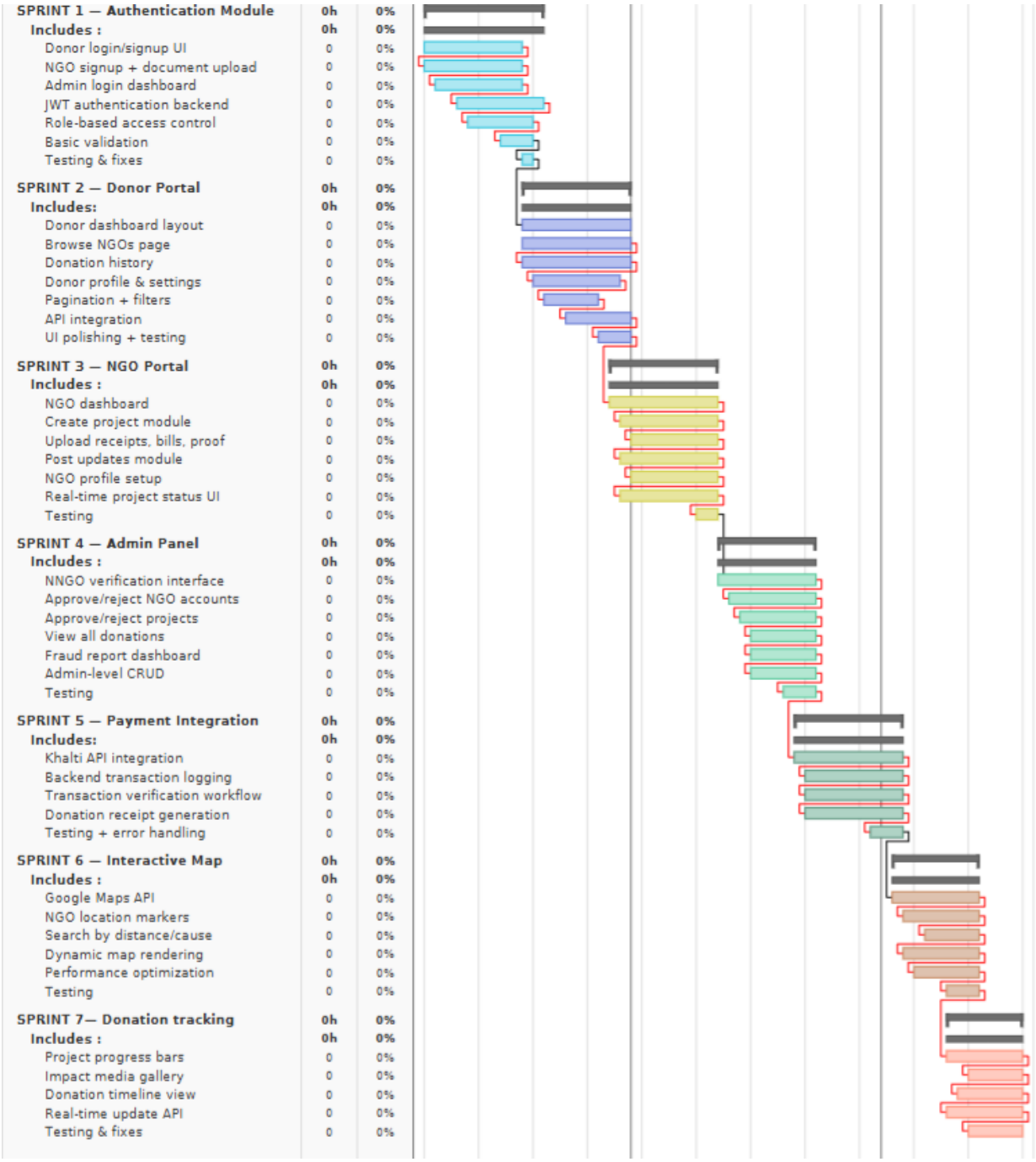

Figure 4 gantt chart development phase

10. Conclusion

This project proposal has articulated the fragmented image of the charitable environment in Nepal as also the necessity of open and centralised Donation Management System. The need to have one digital platform is demonstrated in the proposal by identifying the main challenges that funders and NGOs must overcome including ineffective reporting procedures, unreliable information, inadequacy of transparency, and imbalanced verification. The suggested system will rebuild donor trust by ensuring NGO profiles are made sure, live tracking, systematically updates can be made, payment is received and the payment is risk free.

The proposal also included the goals, objectives, anticipated results, methodology, risks and milestones, required towards the successful completion of the project. Agile Scrum model has also guaranteed that one developer is involved in iterative development, and constant feedback and manageable workloads. The conclusion is that this plan has offered a good ground to build a working, transparent and trustworthy contribution management system that may call up the culture of giving in Nepal and to improve accountability.

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